

A HYBRID DEEP LEARNING-BASED APPROACH FOR NETWORK ANOMALY DETECTION

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FEBRUARY 2021

COHORT 6

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INTRODUCTION

▪ Anomaly

- ❖ “An observation or subset of observations which appears to be inconsistent with the remainder of that set of data” -Barnett & Lewis
- ❖ Outlier/Novelty
- ❖ Network Anomaly/Cyber Attacks

▪ Anomaly Detection

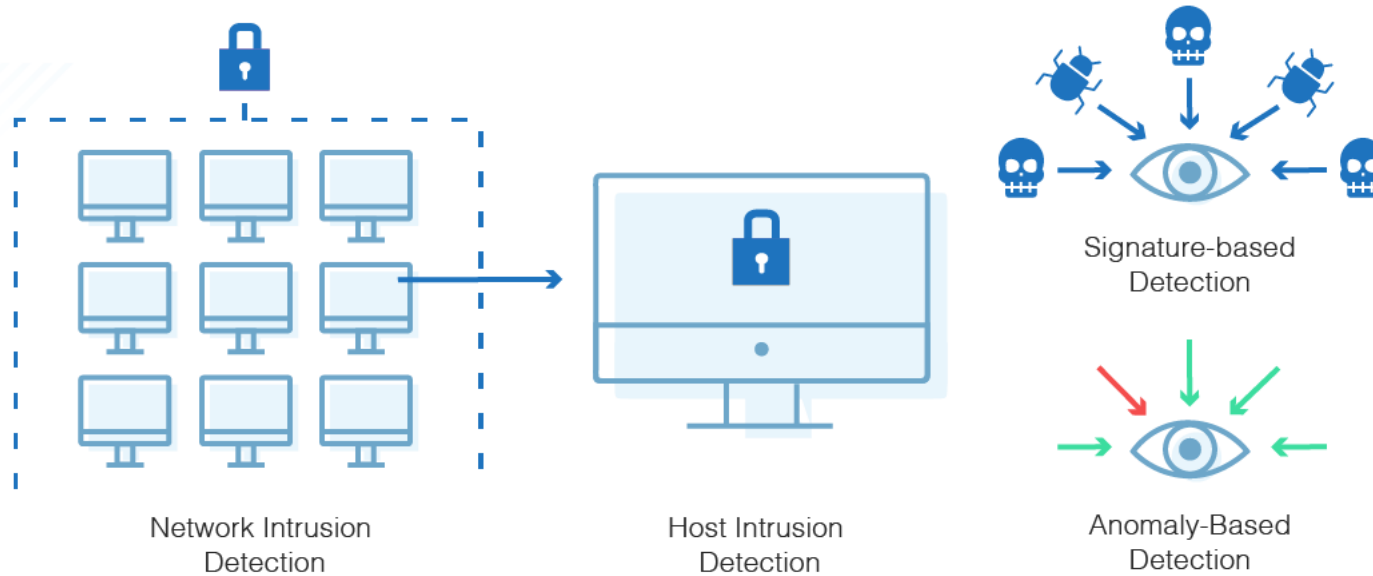
- ❖ Industrial Applications & Socio-Economic Impacts
 - **Intrusion Detection**:-Unauthorized access to computer networks
 - **Fraud Detection**:- Fraudulent bank transactions or loan applications
 - **Activity monitoring**:- Suspicious equity trades or mobile frauds
 - **Network Performance**- performance monitoring to detect bottlenecks
 - **Defect Detection**:-Fault detection in manufacturing units

▪ Intrusion Detection System

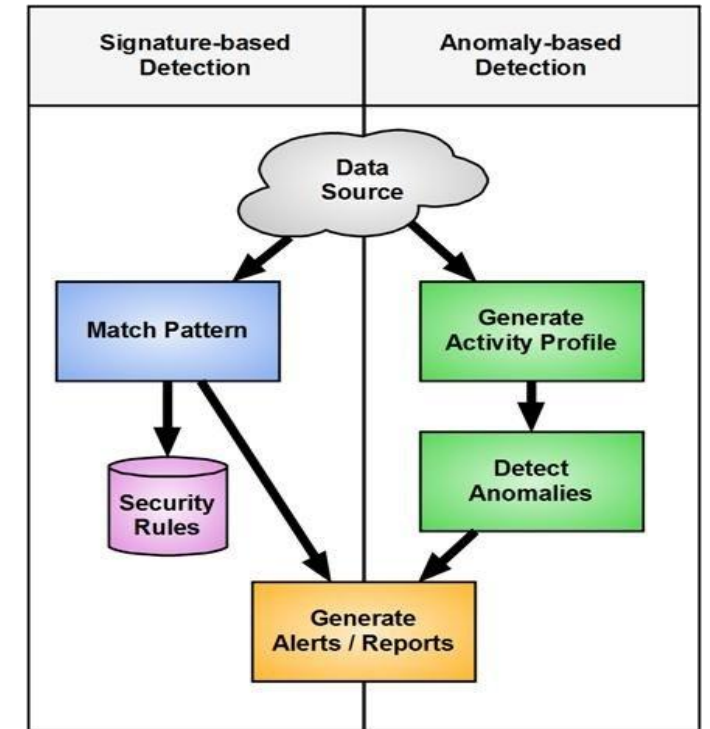
- ❖ Network-Based IDS
- ❖ Host - Based IDS
- ❖ Signature Based Detection
- ❖ Anomaly Based Detection

INTRUSION DETECTION SYSTEM

Intrusion Detection System



IDS, (2019) DNSstuff



PROBLEM STATEMENT

- The average cost of malware attacks and data breaches is forecasted to reach 6.5 trillion dollars annually by 2021. (hackmageddon.com)
- The top 10 cyberattacks in 2019 include 39.3% Malware and 12% Unknown attacks (PurPleSec)
- The major challenges faced by the real-world NIDS are:
 - ❖ Unavailability of labeled and balanced datasets,
 - ❖ High dimensionality
 - ❖ Heterogeneous and nonlinear attack characteristics
 - ❖ Identifying novel attacks.
- Anomaly detection as a binary classification problem -Normal or Attacks
- Network anomaly detection from a multi-attack class perspective – very few works.
 - ❖ Probe,U2R and R2L Attacks- Low Detection Rates.
 - ❖ DoS attacks are easily identifiable.
- The real challenge is identifying novel threats and categorizing them as appropriate attack type based on their attributes

AIM & OBJECTIVE

The main **aim** of this research is to develop a hybrid deep learning-based model for network anomaly detection incorporating the self-learning characteristics of unsupervised techniques and the high predictive power of deep neural networks. The goal of this research is to propose a Network Intrusion Detection Systems more cost-effective and resilient to zero-day attacks by detecting unforeseen anomalies without requiring prior knowledge of network attacks and expensive labeled datasets.

The research **objectives** are formulated based on the aim of this project as follows:

- To preprocess the data for exploratory data analysis and visualization.
- To evaluate deep learning-based feature selection methods for optimal performance.
- To cluster the unlabeled network data patterns using unsupervised machine learning techniques.
- To classify the self-learned patterns using deep neural networks.
- To evaluate the performance of the proposed hybrid anomaly detection model.

LITERATURE REVIEW

Anomaly Detection

- ❖ Probability and Mathematical methods(Tukey,1960)
- ❖ Statistical Outlier Detection(1970s)
- ❖ Rule –based Algorithms(Early 2000's)
- ❖ Machine Learning and Deep Learning (Beyond 2009)

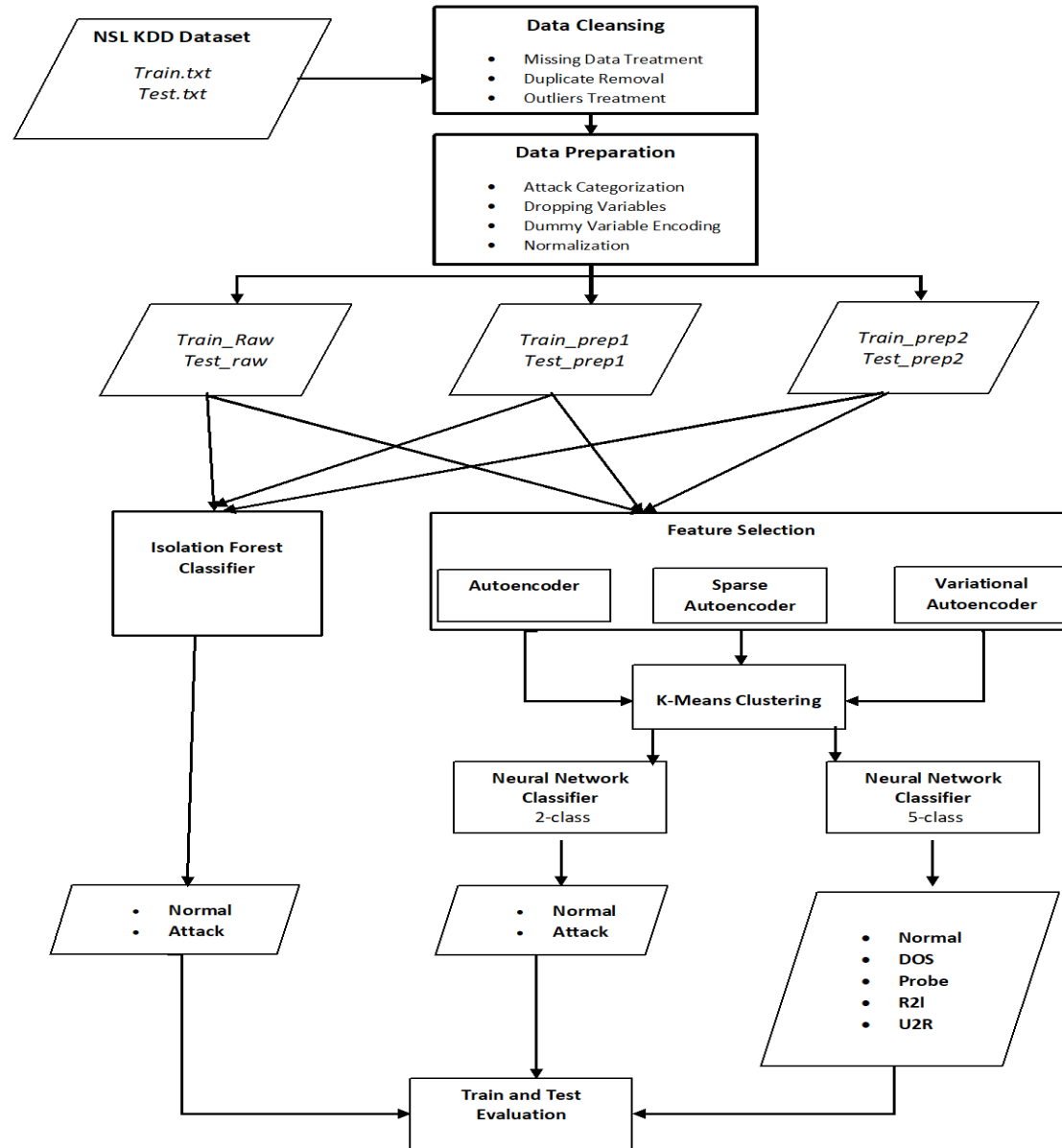
Network Anomaly Detection

- ❖ Statistical Methods
- ❖ Supervised :- RF, DT, CART
- ❖ Unsupervised:- IF, K-means, LOF
- ❖ Feature Ensemble Approach
- ❖ Auto Encoder Based :- AE, VAE, Stacked AE
- ❖ DNN, RNN, LSTM, BLSTM
- ❖ Hybrid Approaches- ML +DL

LITERATURE REVIEW

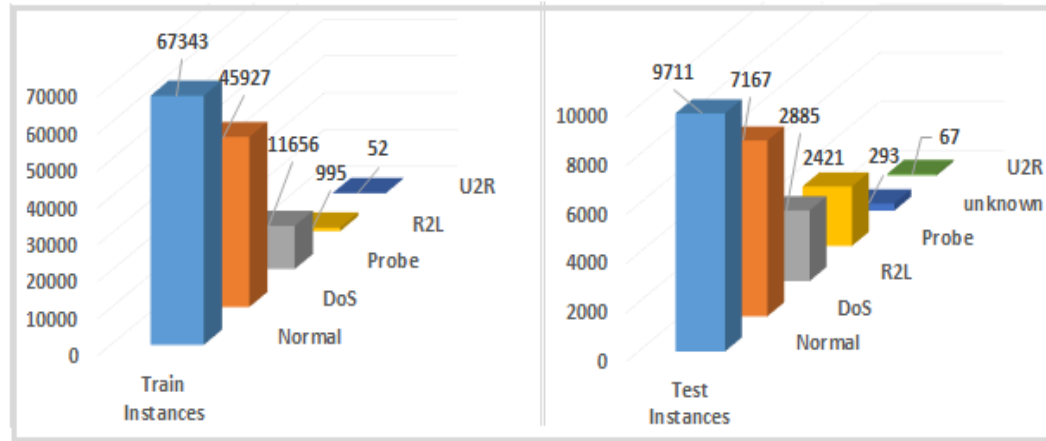
Anomaly detection Approach	PROS	CONS
Statistical Methods	High accuracy on small datasets	Curse of dimensionality & not suitable for multivariate analysis
Supervised ML	High accuracy	Requires Label
Unsupervised ML	High Accuracy , Labeled Data Not Required, Responsive to unknown threats	High False Alarm Rate
Ensemble Approach	High Accuracy	High Computational Overhead and Time
Auto Encoder Based	Optimal Feature Representation and Dimensionality Reduction	Classification accuracy dependent on selected threshold value and High Computational Time
Deep Learning	High Accuracy and ability to deal with Nonlinearity	High computational parameters
Hybrid (ML +DL)	High Performance Improvement	High Computational Parameters(can be optimized)

METHODOLOGY



METHODOLOGY

Data Preprocessing



DATA PREPARATION	Dropped Variables		Dummy Variable		Normalization
	Low Variance	High Correlation	Categorical	Ordinal	
kdd_train_raw	✓	✗	✓	✗	✓
kdd_train_prep1	✓	✓	✓	✓	✓
kdd_train_prep2	✓	✓	✓	✗	✓
kdd_test_raw	✓	✗	✓	✗	✓
kdd_test_prep1	✓	✓	✓	✓	✓
kdd_test_prep2	✓	✓	✓	✗	✓

Hardware Resources

- GPU enabled cloud platform using Nimblebox for performing Deep Learning

Software Resources

- Dataset:- NSL KDD Dataset from Kaggle (<https://www.kaggle.com/anushonkar/network-anomaly-detection?select=Network+Anomaly+Detection.docx>)
- Programming:- Python
- Libraries:- Pandas, Numpy, Scikitlearn, TensorFlow, Keras, Seaborn, Matplotlib

ANALYSIS

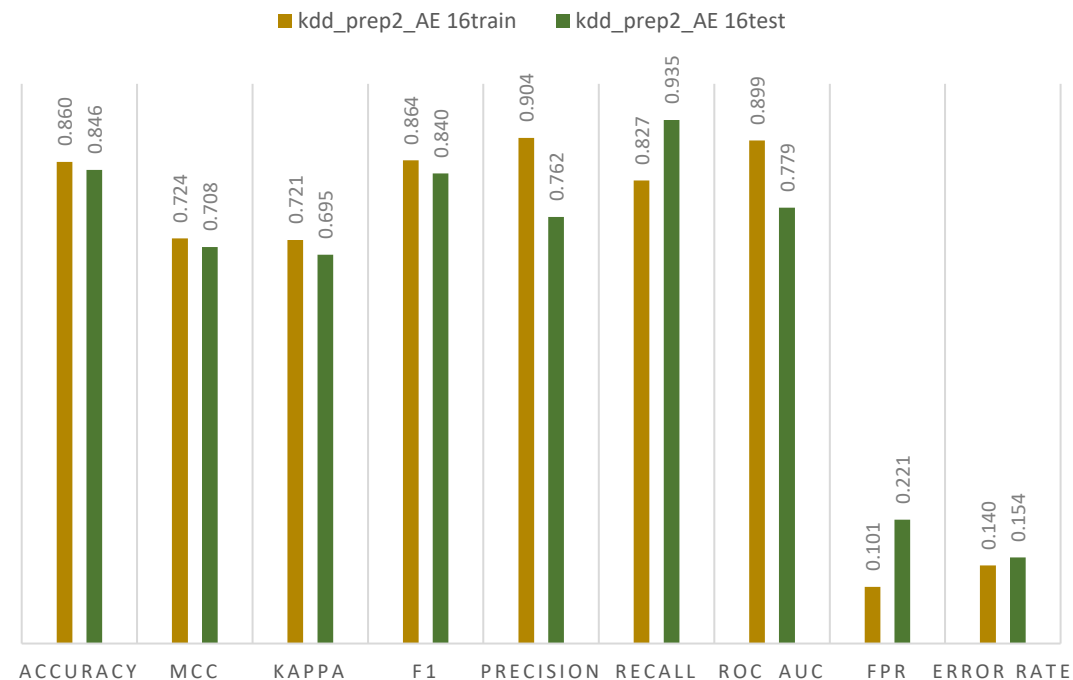
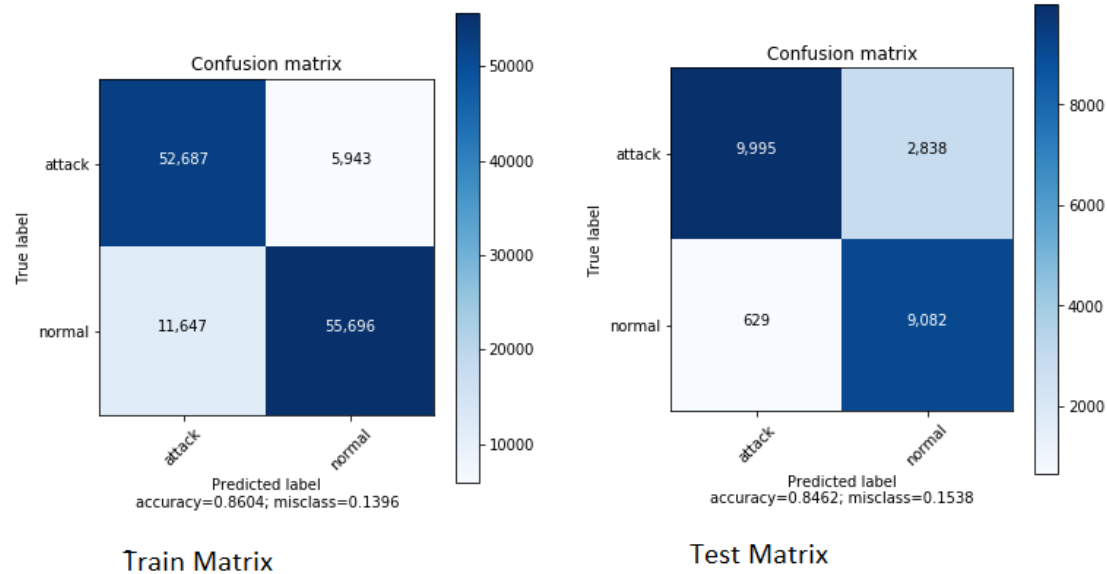
Bivariate Analysis-Insights

- The Normal and Attack distribution in Train and Test datasets are not proportional.
- ***'is_host_login'*** and ***'num_outbound_cmds'*** -Nil variance.
- A well distinguishing pattern in the frequency distribution of basic and content variables for the Normal and Attack types.
- A significant difference in the data distribution of Normal and DoS traffic types
- Similar patterns in the test data distribution of Normal, R2I, and U2R traffic types for Time and Host related attributes,
- Similar distribution patterns in DoS and Probe Attack types for Time and Host related attributes for test data.
- High absolute correlation observed among certain attributes.

RESULTS

Binary Classification

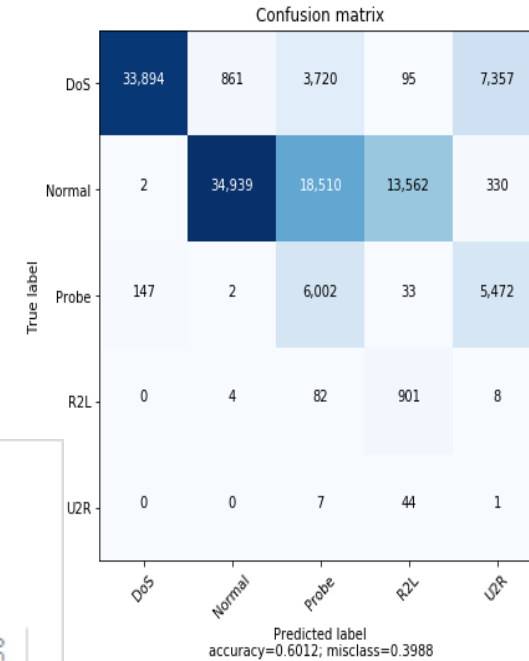
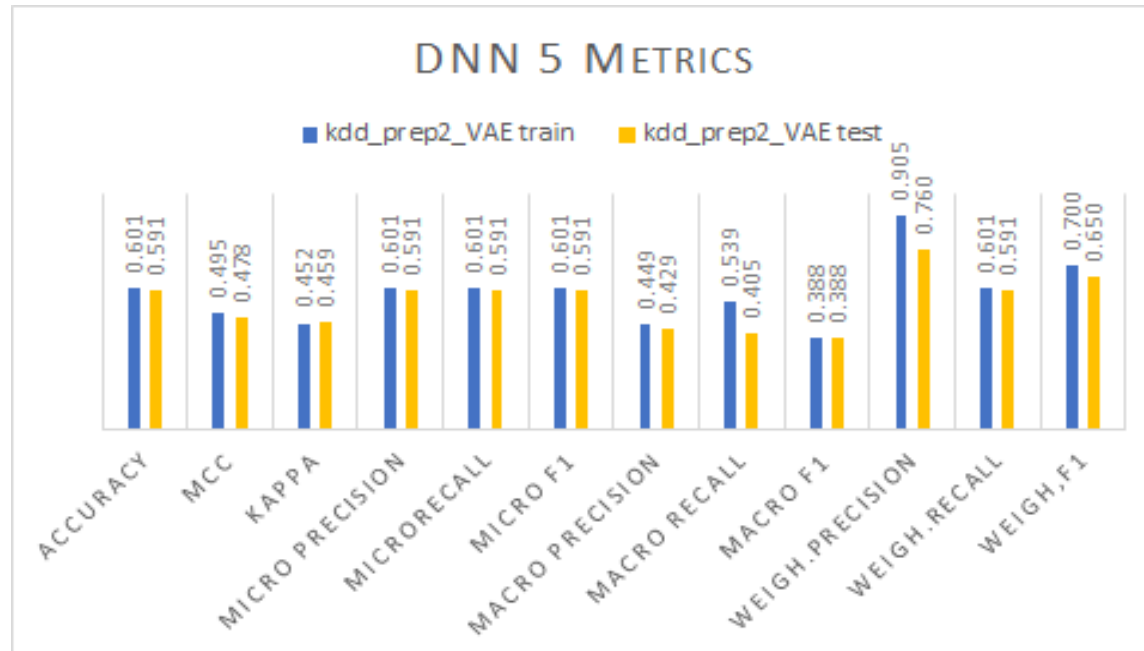
- ❖ kdd_prep2_AE_16
- ❖ Accuracy- 0.846
- ❖ Recall- 0.935
- ❖ F1-score-0.840
- ❖ Run Time -23.17 sec



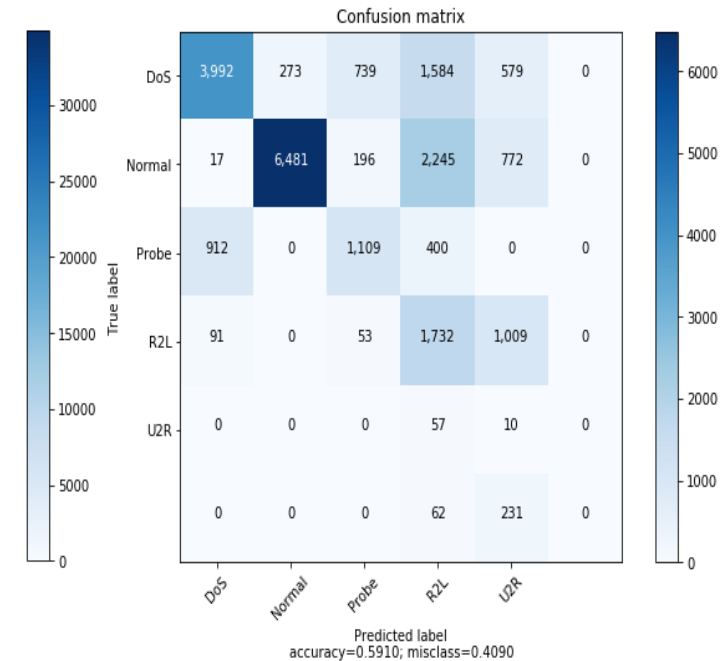
RESULTS CONTD.....

Multi-Class Attack Classification

- ❖ kdd_prep2_VAE
- ❖ Accuracy- 0.59
- ❖ Weight.F1-score-0.65
- ❖ Run Time -33sec



Train



Test

DISCUSSIONS

- kdd_prep2 performed well in 2class and 5 class.
- 20% performance improvement in recall and accuracy with AE FS, K-means, and DNN 2 class classification over Isolation Forest.
- AE and VAE better performed with the nonlinear activation functions 'elu' and 'tanh'.
- The moderate performance exhibited by sparse autoencoder may be suspected to the L1 regularizer used in hidden layers.
- With VAE FS, K-means, and DNN 5 class classification achieved lower values of false alarm rate on different attack types with a relatively lesser computation time, but at the cost of the detection rate for Normal and DoS.
- Normal traffic being misclassified majorly as R2L and U2R attack types
- DoS traffics are being misinterpreted as Probe
- Unknown attacks (DOS) missed in 2 class
- Unknown attacks(DOS) misinterpreted as R2L and U2R in 5 class.

CONCLUSIONS, FUTURE WORKS AND LIMITATIONS

- The results for anomaly Detection is promising enough and on par with the state-of-the-art semi supervised classifiers performance.
- The classification time is relatively less which makes it recommendable for real-time anomaly detection
- The Hybrid approach combining ML and DL techniques achieved satisfactory outputs for anomaly detection.
- A cost-effective and time-efficient approach for Network Intrusion Detection.
- Limitations:-
 - Average performance on attack categorization and poor response to novel attack
 - Static Data
- Further works include:
 - Other feature selection approaches :- Stacked autoencoders and non-symmetric autoencoders
 - Semi-supervised techniques by extracting Normal network traffic

THANK YOU